AI Powered Smart Water Management System Using IoT and AI

SDG 9: Industry, Innovation and Infrastructure

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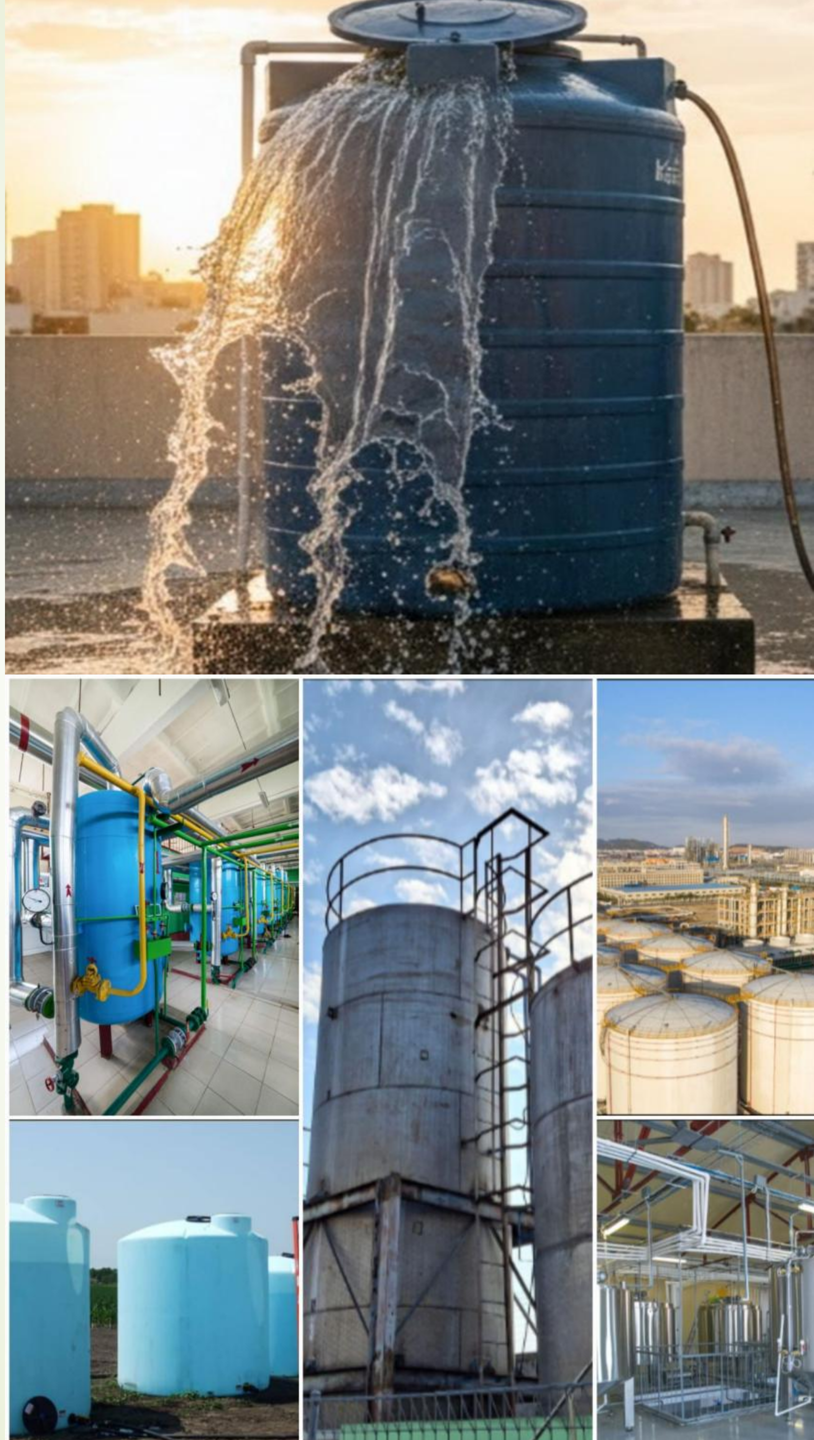
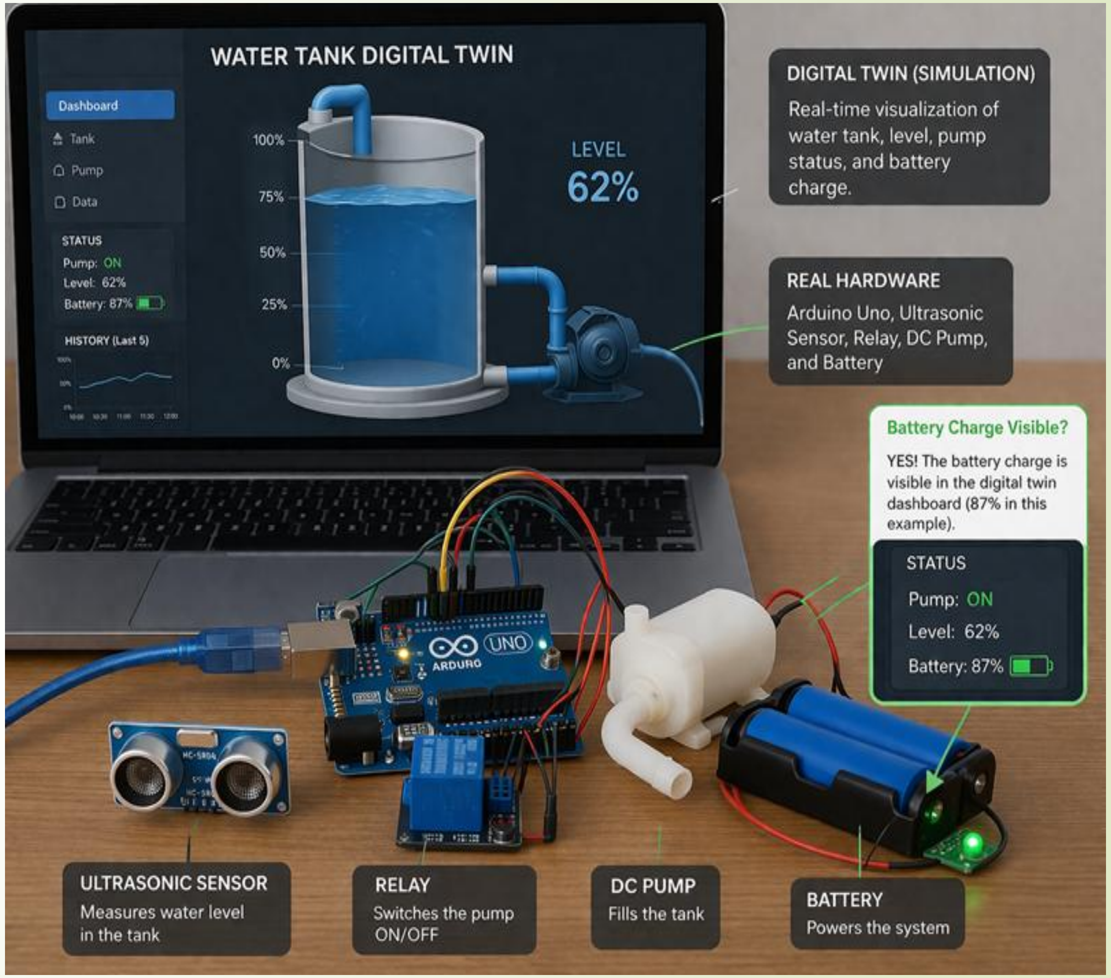
Problem Statement

- Water management in industries is a critical challenge due to inefficient monitoring and lack of automation.
- **Existing systems** use **contact-based sensors (float sensors)**
  - Require physical contact with water
  - Prone to corrosion and damage
- **Manual monitoring** issues
  - Workers must climb large tanks → unsafe and time-consuming
- In **chemical industries**
  - Tanks cannot be opened due to toxic gases or vapors
  - Direct monitoring is risky
- **Lack of automation**
  - Leads to inefficiency, delays, and human errors
  - Causes overflow, water wastage, and pump damage



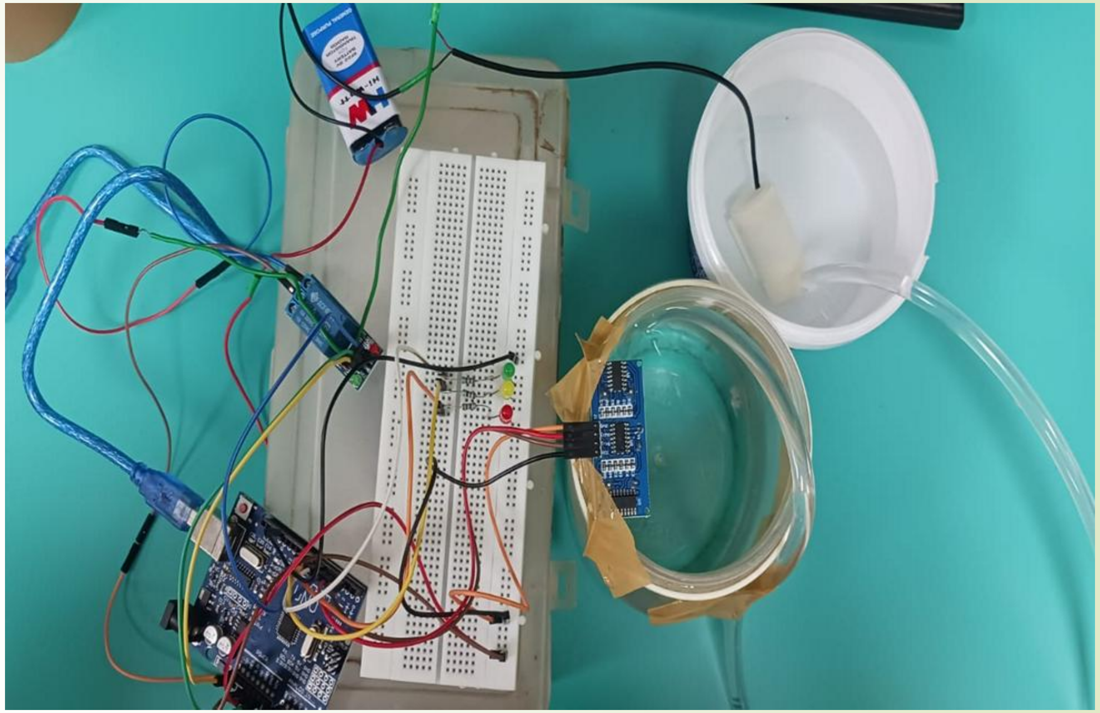
Objectives

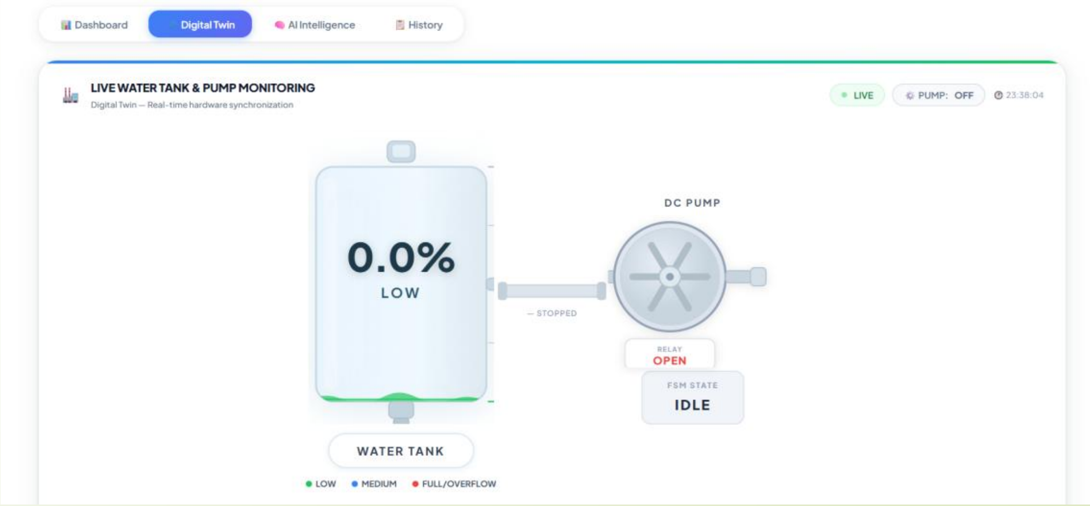
- To monitor water levels in real-time using sensors
  • Continuously track tank levels without manual effort
- To implement non-contact ultrasonic sensing for safety
  • Avoids corrosion and ensures long-term durability
- To integrate Gemini API for intelligent analysis
  • Provides insights and usage pattern understanding
- To automate pump ON/OFF operations using Arduino + Relay
  • Ensures reliable and fast control of the pump
- To prevent overflow and dry running
  • Protects water resources and prevents motor damage
- To improve efficiency and reduce operational cost
  • Optimizes energy and water usage
- To provide a scalable industrial solution
  • Can be extended to industrial-grade systems



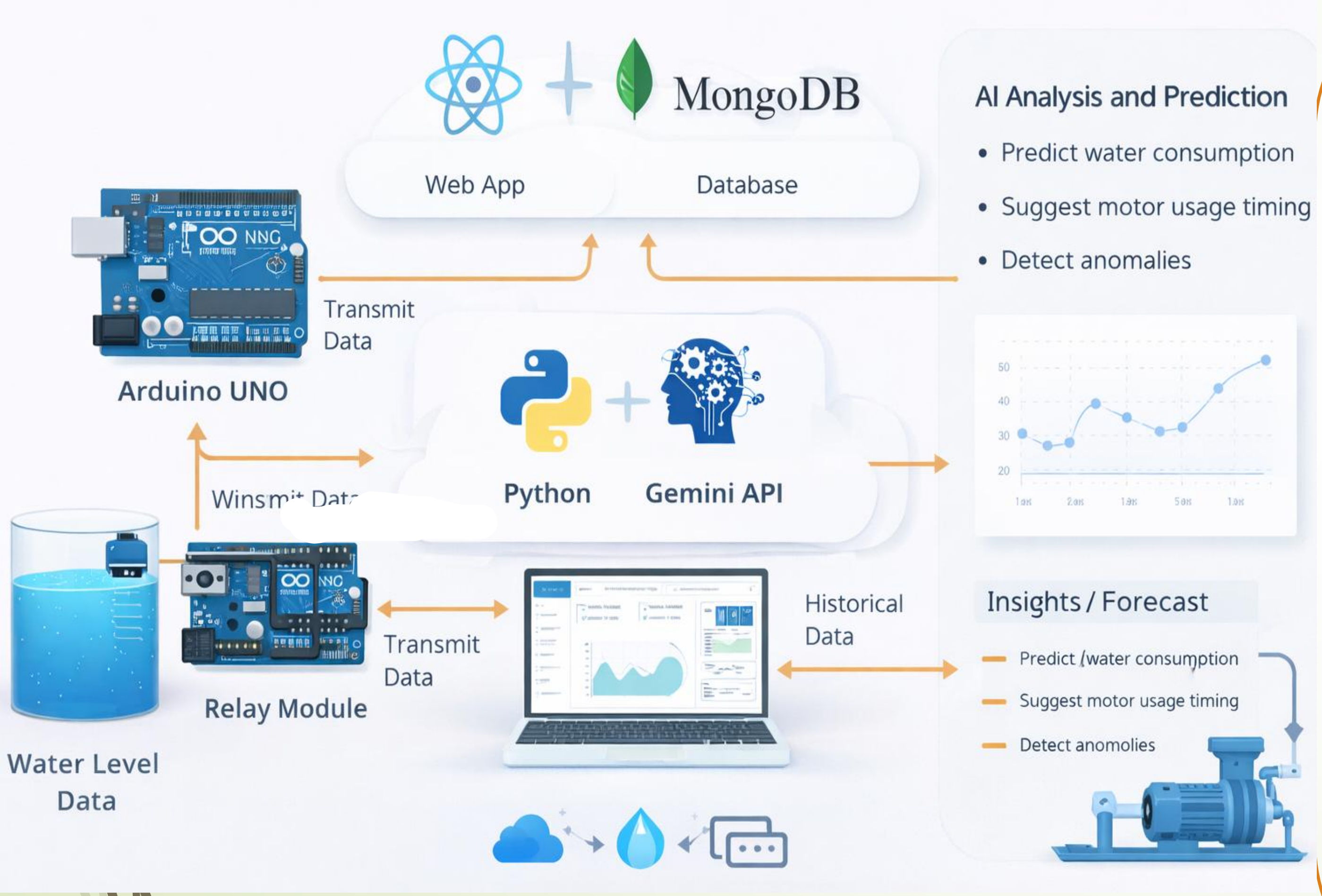
Proposed Solution

- We propose an AI-powered automated water management system for industries.
- Uses ultrasonic sensor for non-contact level measurement
  • Provides accurate readings without physical contact
- Uses Gemini API for intelligent analysis
  • Analyses water usage patterns and gives predictive insights
- Automates pump operations using Arduino and Relay
  • Relay acts as a switch to control high-power pump safely
- Provides real-time monitoring via IoT dashboard
  • Displays live data, alerts, and system status
- Works safely in hazardous environments
  • Suitable for chemical and industrial setups





System Architecture



➤ **Step-by-Step Explanation**

- **Data Collection:**
  Water level sensors continuously measure the water level in the tank
- **Data Transmission:**
  Microcontroller (Arduino/ESP32) collects and transmits data
- **Data Processing:**
  The collected data is filtered and formatted
- **AI Analysis (Gemini API):**
  Gemini API analyzes the data and predicts usage patterns
- **Decision Making :**
  Pump ON/OFF decision is taken by Arduino using predefined threshold logic
- **Control Mechanism:**
  Relay module acts as a switch to control the pump
- **Output:**
  Pump is controlled automatically and alerts are sent to users

Methodology

➤ Sensor measures water level continuously
• Sends real-time data at regular intervals
➤ Data sent to Arduino/ESP32
• Microcontroller collects and processes the data
➤ Data processed and forwarded to AI system
• Data is filtered and prepared for analysis
➤ Gemini API analyzes usage patterns
• Provides intelligent insights and predictions
➤ System decides pump ON/OFF
• Based on threshold values using Arduino logic
➤ Relay controls the pump
• Acts as an interface between Arduino and motor
➤ Alerts and automation executed
• Notifications sent and pump controlled automatically

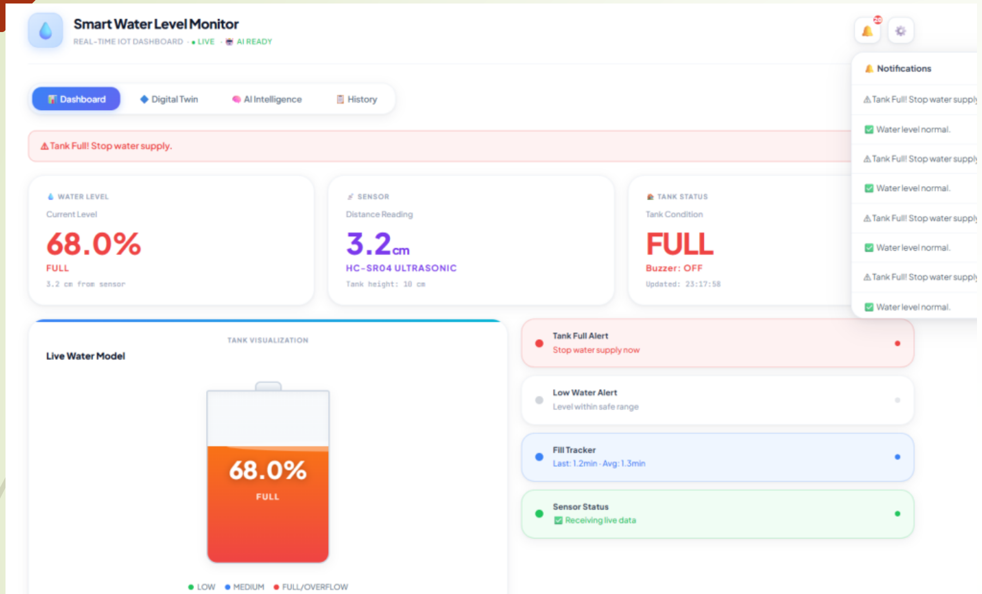
AUTOMATION FEATURES:

• **Automatic tank filling**
→ Pump starts and stops automatically based on water level
• **Smart start/stop logic**
→ Avoids frequent switching, increasing pump life
• **Overflow prevention**
→ Stops pump before tank becomes full
• **Remote pump control**
→ Allows control from dashboard or mobile

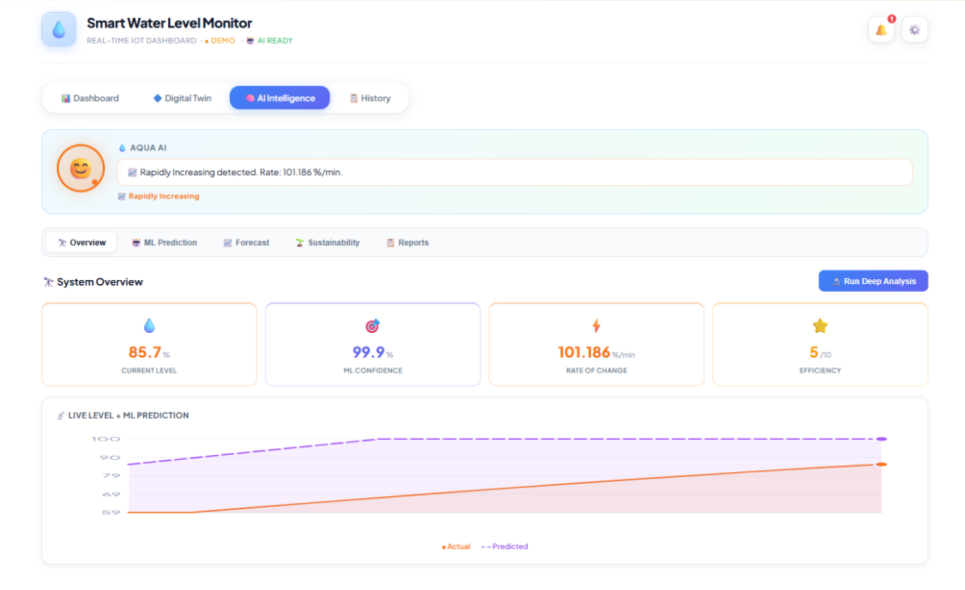
Tools and Technologies

➤ **Hardware**
• Ultrasonic Sensor → measures water level without contact
• Arduino UNO → controls system and processes data
➤ **Software**
• Python → data processing and AI integration
➤ **AI**
• Gemini API → performs intelligent analysis and prediction
➤ **IoT**
• MQTT / WebSocket / REST API → enable real-time communication
➤ **Dashboard**
• React / Web Interface → displays system data and alerts

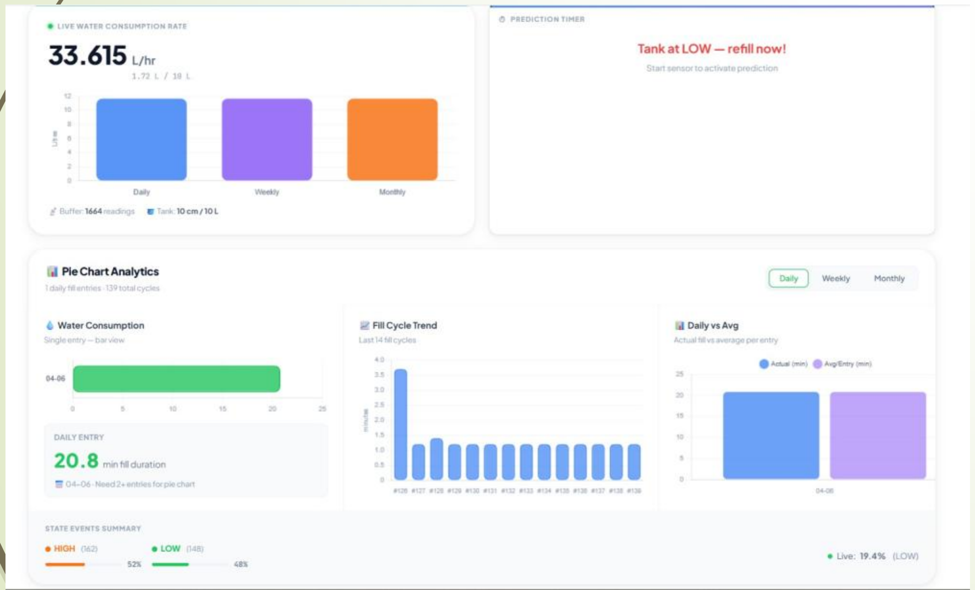
Result



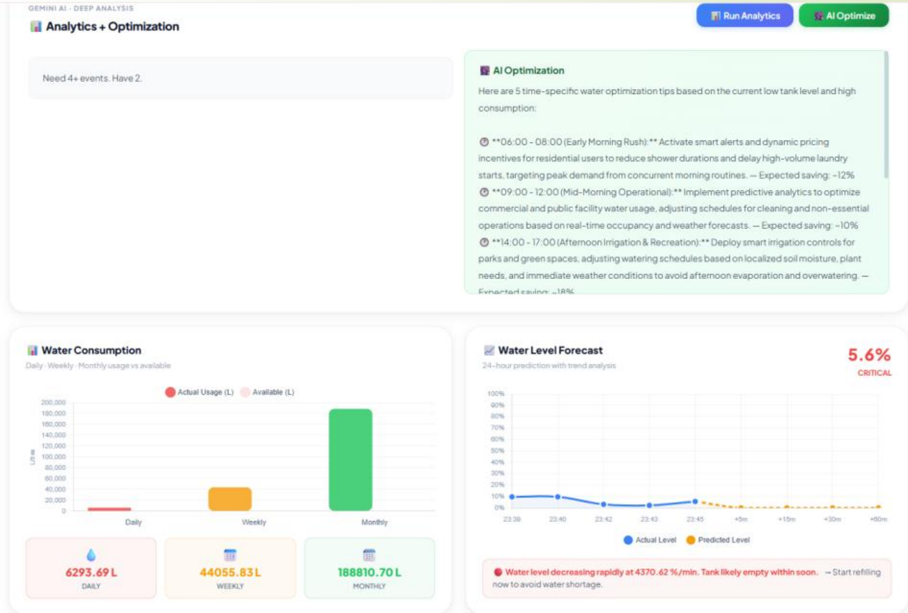
Smart Water Level Monitor
REAL-TIME IOT DASHBOARD · LIVE · AI READY
Dashboard
Digital Twin
AI Intelligence
History
Notifications
Tank Full! Stop water supply.
Water level normal.
WATER LEVEL
Current Level
68.0%
FULL
SENSOR
Distance Reading
3.2 cm
HC-SR04 ULTRASONIC
TANK STATUS
Tank Condition
FULL
Buzzer: OFF
Live Water Model
68.0%
FULL
Tank Full Alert
Stop water supply now
Low Water Alert
Level within safe range
Fill Tracker
Last: 1.2min · Avg: 1.3min
Sensor Status
Receiving live data
LOW
MEDIUM
FULL/OVERFLOW



Smart Water Level Monitor
REAL-TIME IOT DASHBOARD · DEMO · AI READY
Dashboard
Digital Twin
AI Intelligence
History
AQUA AI
Rapidly increasing detected. Rate: 101.186 %/min.
Rapidly Increasing
Overview
ML Prediction
Forecast
Sustainability
Reports
System Overview
Run Deep Analysis
85.7 %
CURRENT LEVEL
99.9 %
ML CONFIDENCE
101.186 %/min
RATE OF CHANGE
5 /10
EFFICIENCY
LIVE LEVEL + ML PREDICTION
Actual
Predicted



LIVE WATER CONSUMPTION RATE
33.615 L/hr
Daily
Weekly
Monthly
PREDICTION TIMER
Tank at LOW — refill now!
Start sensor to activate prediction
Pie Chart Analytics
Water Consumption
Fill Cycle Trend
Daily vs Avg
DAILY ENTRY
20.8 min fill duration
STATE EVENTS SUMMARY
HIGH
LOW
Live: 19.4% (LOW)



Analytics + Optimization
Run Analytics
AI Optimize
Need 4+ events. Have 2.
AI Optimization
Here are 5 time-specific water optimization tips based on the current low tank level and high consumption:
Water Consumption
Water Level Forecast
5.6%
CRITICAL
6293.69 L
DAILY
44055.83 L
WEEKLY
188810.70 L
MONTHLY
Actual Level
Predicted Level

Advantage / Limitations

- **Advantages**

• Non-contact sensing → no corrosion and longer life
• Safe for hazardous environments like chemical industries
• Reduces manual effort and improves safety
• Prevents water wastage and overflow
• Improves efficiency through automation
• Supports Industry 4.0 concepts

- **Limitations**

• HC-SR04 limited range (2–4 meters)
• Accuracy affected by steam, dust, and temperature
• Not suitable for very large industrial tanks
• Requires industrial-grade sensors for large-scale implementation

Conclusion

- This project provides an intelligent and automated solution for industrial water management.
By integrating IoT and AI, the system improves efficiency, reduces water wastage, and ensures safe and reliable operation.
It minimizes human intervention and can be scaled for real-world industrial applications.

Future Scope

- Use industrial-grade ultrasonic or radar sensors
  - Improve accuracy and support large industrial tanks
- Integration of Raspberry Pi + Camera Module
  - Enables visual monitoring and advanced detection
- **Industry Applications of Camera**
- • Oil & Gas → detect leakage and contamination
  • Chemical Industry → detect color change and foam formation
  • Water Treatment → check water clarity and impurities
  • Food Industry → monitor quality and hygiene
- Implement advanced AI models (LSTM, Reinforcement Learning)
  - Improve prediction accuracy and decision-making
- Integration with SCADA systems
  - Enable centralized industrial control
- Full Industry 4.0 automation
  - Smart, connected, and fully automated industrial system

Thank you !